Marketing Funnel & Lead Conversion Analysis

Bank Direct Marketing Campaign — Business Analytics Report

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Role	Data Analytics Intern
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GitHub Repository	github.com/priyam-10/FUTURE_DS_03
Dataset	UCI Bank Marketing Dataset (45,211 records)

Confidential — Prepared for Internship Submission & Portfolio Showcase

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1. Executive Summary

This report presents a comprehensive marketing funnel and lead conversion analysis of a Portuguese retail bank's direct marketing campaigns, based on the widely referenced UCI Bank Marketing dataset containing 45,211 customer interactions across 17 attributes. The objective of the study was to quantify campaign effectiveness, identify the customer segments most likely to subscribe to a term deposit, and translate those observations into actionable marketing recommendations.

The analysis reveals that the overall conversion rate is relatively low — a typical characteristic of cold-outreach financial campaigns — yet significant performance differences exist across contact channels, calendar months, and customer demographics. Customers contacted via cellular channels, those with higher account balances, students and retirees, and customers with a positive outcome in a previous campaign all demonstrate materially higher subscription rates. These findings support a shift from mass outreach toward a data-driven, segment-first targeting strategy that can increase conversion while reducing per-lead marketing cost.

2. Project Overview

The Marketing Funnel & Lead Conversion Analysis project was developed as part of a Data Analytics Internship. Using Python, Pandas, Matplotlib and Seaborn in a Jupyter Notebook environment, the study examines how customer demographics, communication channels, campaign timing and prior engagement influence the likelihood of subscribing to a term deposit. The complete implementation is available in the public repository **github.com/priyam-10/FUTURE_DS_03**.

3. Problem Statement

Banks invest significant resources into direct marketing campaigns, yet a substantial portion of those interactions do not result in a subscription. Without a clear understanding of which channels, time windows and customer segments actually convert, marketing spend is diluted and customer experience suffers due to over-contact. This project addresses the question: *which levers most strongly influence customer conversion, and how should the bank reallocate effort to improve return on marketing investment?*

4. Project Objectives

- Measure the overall subscription (conversion) rate of the campaign.
- Compare the effectiveness of cellular vs telephone contact methods.
- Identify the calendar months that generate the highest conversion.
- Segment customers by job, education, marital status, age and balance.
- Evaluate the influence of the previous campaign's outcome on future conversion.
- Translate analytical findings into practical business recommendations.

5. Dataset Overview

The analysis uses the **Bank Marketing Dataset** from the UCI Machine Learning Repository, collected from a Portuguese retail bank's direct marketing campaigns conducted primarily via phone.

Attribute	Value
Source	UCI Machine Learning Repository
Records	45,211
Features	17
Target Variable	y (term deposit subscription: yes / no)
Domain	Retail Banking — Direct Marketing

Variable Categories

- **Demographics:** age, job, marital status, education.
- **Financial:** account balance, housing loan, personal loan, credit default flag.
- **Campaign contact:** contact type, day, month, call duration, number of contacts.
- **Previous campaign:** days since last contact (pdays), previous contacts, outcome.
- **Target:** whether the customer subscribed to a term deposit.

6. Tools and Technologies Used

Technology	Purpose
Python	Core analysis language
Pandas	Data cleaning and manipulation
NumPy	Numerical computation
Matplotlib	Data visualization
Seaborn	Statistical visualization
Jupyter Notebook	Interactive development environment
Git & GitHub	Version control and portfolio hosting

7. Methodology

The project followed a structured analytics workflow that mirrors industry practice for marketing analytics engagements:

7.1 Data Collection

The raw dataset was obtained from the UCI Machine Learning Repository and loaded into a Pandas DataFrame for inspection.

7.2 Data Cleaning

Column data types were validated, categorical fields standardised, and 'unknown' values were reviewed. No duplicate records were found. The target variable 'y' was encoded into a binary form for conversion-rate calculations.

7.3 Exploratory Data Analysis (EDA)

Univariate and bivariate analyses were carried out to understand the distribution of customers across demographic and campaign attributes, and to identify preliminary relationships with the target variable.

7.4 Conversion Analysis

Group-wise subscription rates were computed against contact method, month, job, education, marital status, age band, balance band and previous campaign outcome.

7.5 Visualization

Matplotlib and Seaborn were used to produce clear, business-oriented visualizations that communicate patterns to non-technical stakeholders.

7.6 Business Insight Generation

Each analytical output was translated into a business interpretation and, where appropriate, an operational recommendation.

8. Business Questions

1. What is the overall customer conversion rate?
2. Which contact method achieved the highest conversion rate?
3. Which month recorded the highest customer conversion rate?
4. Which customer job categories have the highest conversion rates?
5. Does education level influence customer conversion?
6. Does marital status affect customer subscription?
7. How does customer age impact conversion rates?
8. Does account balance influence customer subscription?
9. How does the previous marketing campaign outcome influence conversion?
10. What recommendations can improve future marketing campaign performance?

9. Exploratory Data Analysis Summary

The dataset is well-structured and largely complete. The customer base skews toward middle-aged working adults, with the 30–50 age band representing the largest share. Most customers were contacted via cellular channels, and campaign volume is heavily concentrated in May, followed by July and August. The target variable is imbalanced: the majority of customers did not subscribe, which is typical for cold-outreach campaigns and shapes the interpretation of every downstream metric.

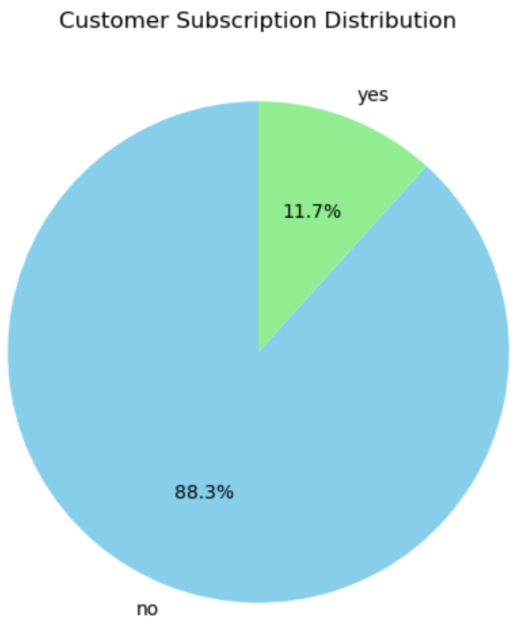

Figure 1: Customer Subscription Distribution

10. Detailed Analysis

10.1 Overall Customer Conversion Rate

Objective

Establish a baseline conversion rate against which all segment-level performance can be compared.

Analysis

The number of customers who subscribed to a term deposit was divided by the total number of customers contacted, producing the overall conversion rate for the campaign.

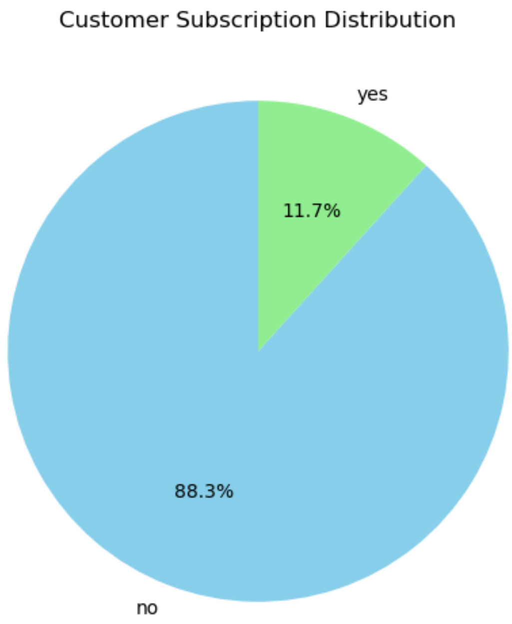

Figure 2: Overall Subscription Rate

Key Observations

- The overall conversion rate is low — the majority of contacted customers did not subscribe.
- The result confirms the class imbalance present in the target variable.
- This baseline becomes the reference point for every segment-level comparison.

Business Interpretation

A low baseline conversion rate is expected for cold outbound financial campaigns. The strategic implication is not that the campaign has failed, but that value is created by identifying and prioritising the sub-segments that convert well above baseline.

10.2 Contact Method Analysis

Objective

Compare the effectiveness of the two dominant contact channels — cellular and telephone — to guide channel investment.

Analysis

Conversion rates were computed for each value of the 'contact' attribute, allowing a direct comparison between cellular, telephone and unknown contact methods.

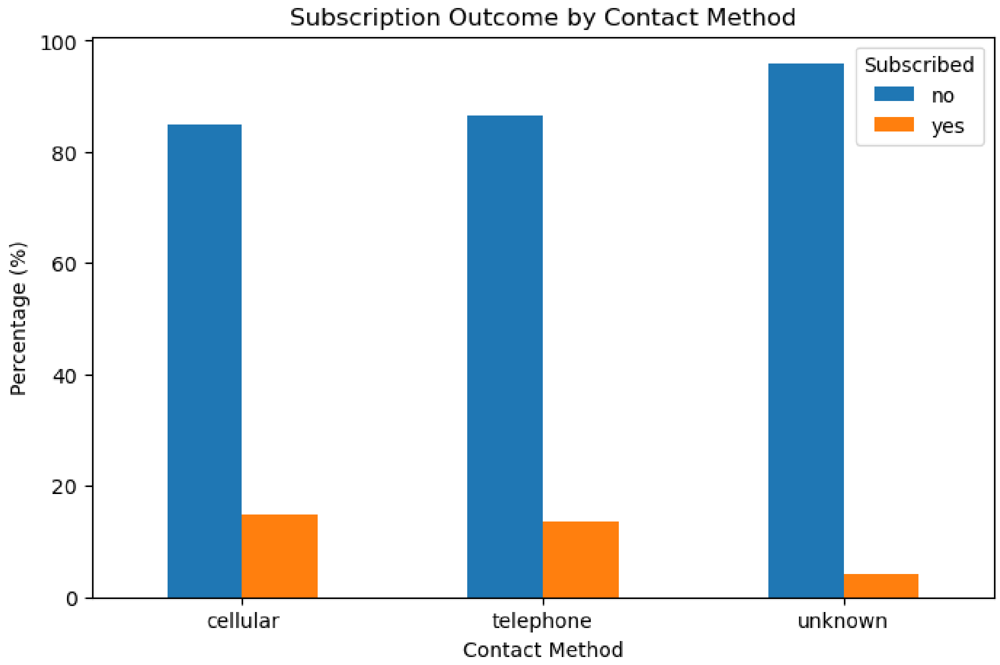

Figure 3: Conversion Rate by Contact Method

Key Observations

- Cellular contact consistently outperforms fixed telephone contact.
- The 'unknown' contact category shows notably weaker conversion.
- Cellular already accounts for the majority of contact volume.

Business Interpretation

Cellular is the strongest performing channel and should remain the default. Investment in improving mobile-first outreach — including SMS follow-ups and consent-based mobile messaging — is likely to yield the highest incremental return.

10.3 Monthly Campaign Performance

Objective

Identify the months in which customers respond most positively so that future campaigns can be scheduled optimally.

Analysis

The subscription rate was computed for each calendar month of contact, and results were ranked to reveal seasonal patterns.

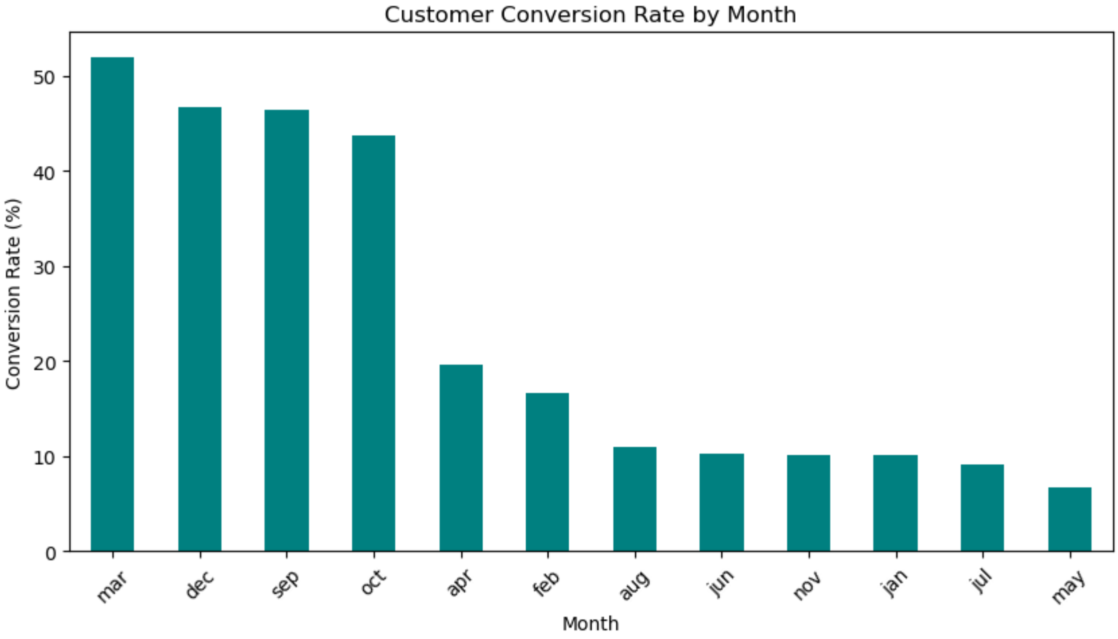

Figure 4: Monthly Conversion Rate

Key Observations

- High-volume months such as May do not always deliver the best conversion rate.
- Lower-volume months (for example March, September, October and December) produce disproportionately high conversion.
- There is a clear mismatch between where volume is spent and where conversion is earned.

Business Interpretation

Reallocating campaign volume from over-saturated months to historically high-converting but under-utilised months should lift overall campaign yield without increasing total contact volume.

10.4 Job Category Analysis

Objective

Understand which occupational segments are most receptive to term deposit offers.

Analysis

Conversion rates were computed for each 'job' category and ranked from highest to lowest.

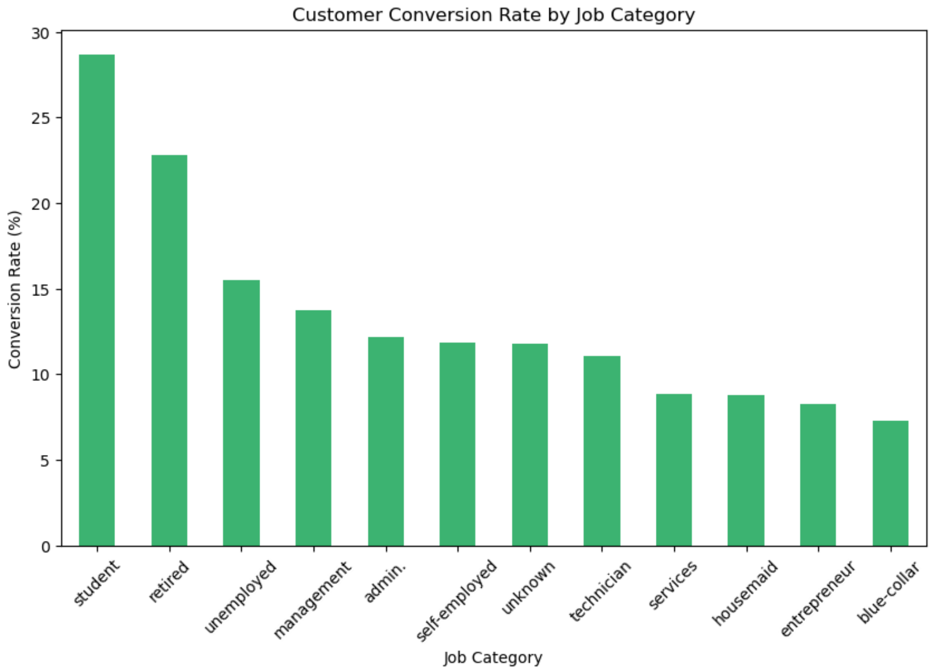

Figure 5: Conversion Rate by Job Category

Key Observations

- Students and retirees show the highest conversion rates.
- Blue-collar workers, entrepreneurs and services staff convert below the campaign average.
- Management and admin roles convert around the campaign average.

Business Interpretation

The bank should design distinct value propositions for high-converting occupations — for example a savings-oriented message for retirees and a first-deposit message for students — rather than relying on a single generic script.

10.5 Education Level Analysis

Objective

Assess whether formal education influences a customer's likelihood to subscribe.

Analysis

Subscription rates were computed for each level of the 'education' attribute.

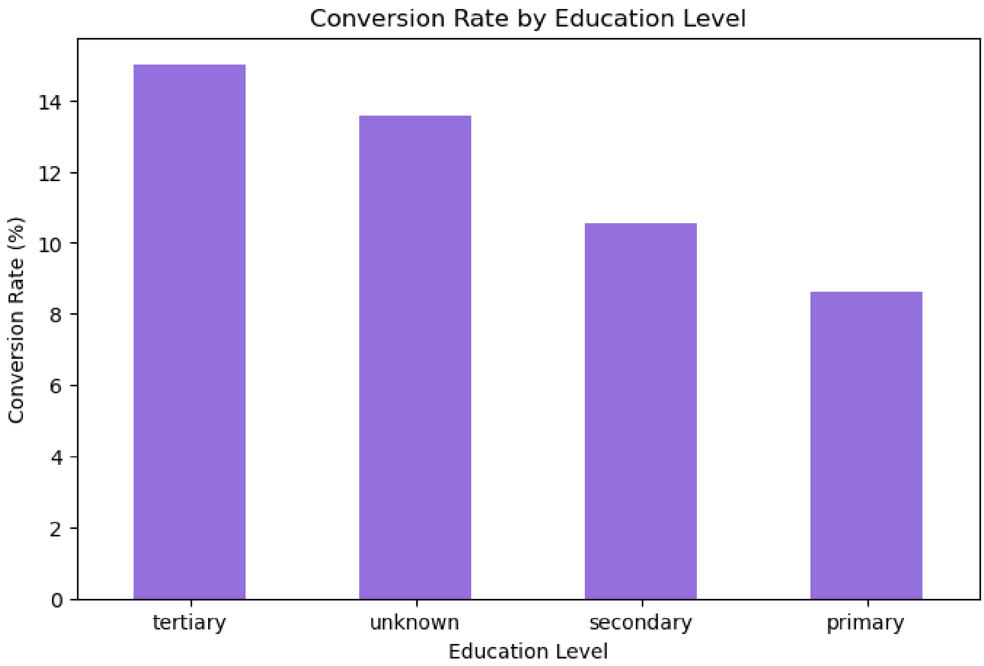

Figure 6: Conversion Rate by Education Level

Key Observations

- Customers with tertiary education convert at the highest rate.
- Primary-educated customers convert at the lowest rate.
- The 'unknown' education category performs moderately.

Business Interpretation

Education level acts as a useful proxy for financial literacy and disposable income. Targeting and messaging should be tiered so that tertiary-educated customers receive product-depth messaging while primary-educated customers receive simplicity- and safety-focused messaging.

10.6 Marital Status Analysis

Objective

Evaluate whether marital status meaningfully influences conversion.

Analysis

The proportion of subscribers was computed for single, married and divorced customers.

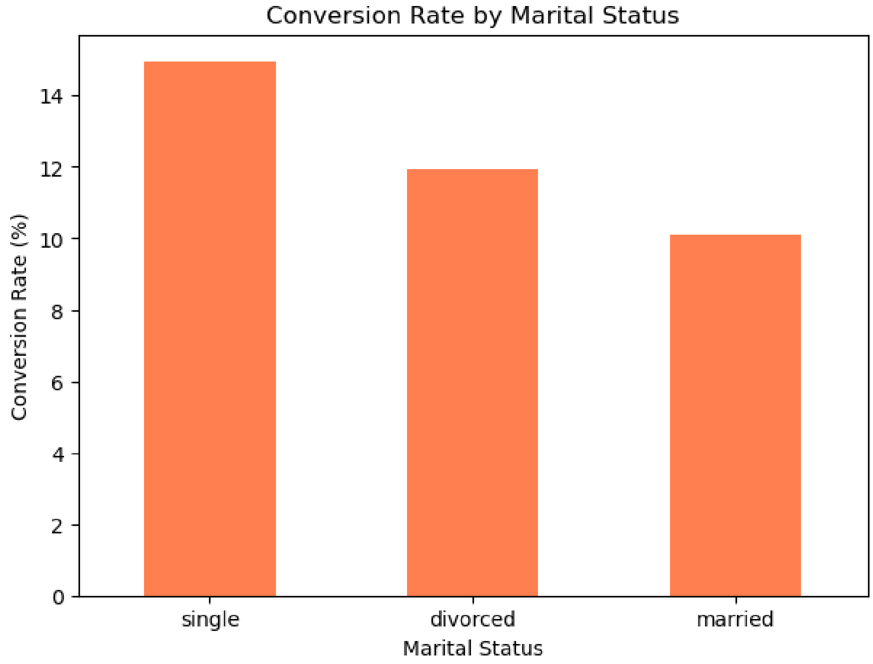

Figure 7: Conversion Rate by Marital Status

Key Observations

- Single customers convert at a higher rate than married or divorced customers.
- Married customers form the largest volume group but convert around the average.
- Divorced customers convert slightly below average.

Business Interpretation

Single customers respond better to individual wealth-building narratives, whereas married customers respond better to family-oriented savings and stability messages. Tailoring the creative narrative to marital status is a low-cost lever for improving conversion.

10.7 Age Group Analysis

Objective

Determine how age influences the probability of subscribing to a term deposit.

Analysis

Age was grouped into interpretable bands (for example under 30, 30–45, 45–60 and 60+) and conversion rates were computed per band.

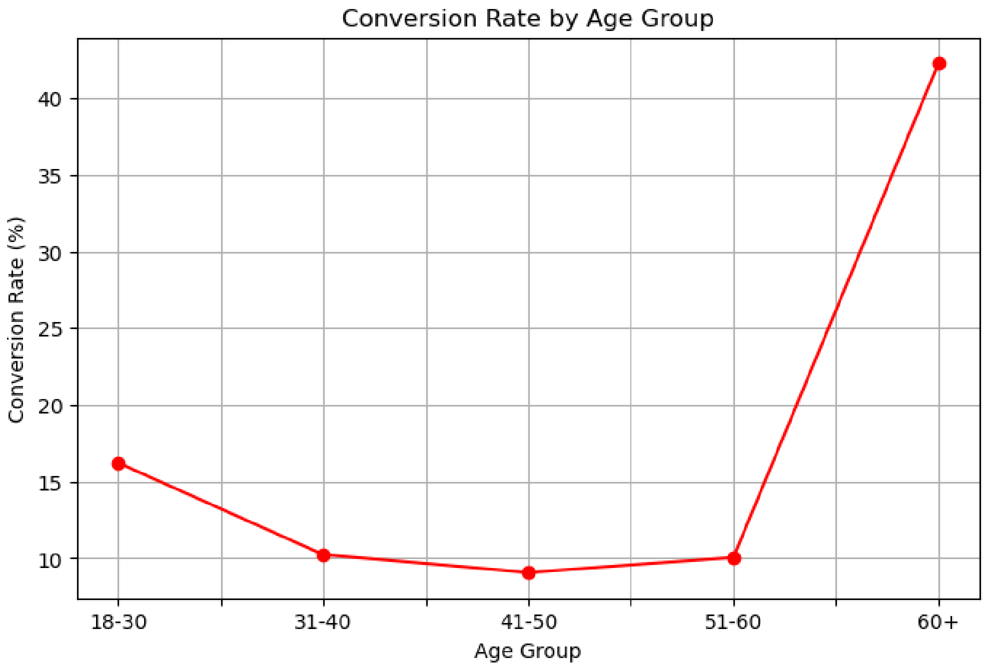

Figure 8: Conversion Rate by Age Group

Key Observations

- The under-30 and 60+ age groups convert at meaningfully higher rates.
- The 30–45 group represents the largest volume but only average conversion.
- Older customers value security and fixed returns — product characteristics of a term deposit.

Business Interpretation

Age-based prioritisation offers one of the clearest opportunities to lift conversion. Focusing on younger first-time savers and older security-seeking customers is likely to outperform undifferentiated outreach.

10.8 Account Balance Analysis

Objective

Test whether existing account balance is a predictor of subscription.

Analysis

Balances were grouped into low, medium and high bands and conversion was computed for each band.

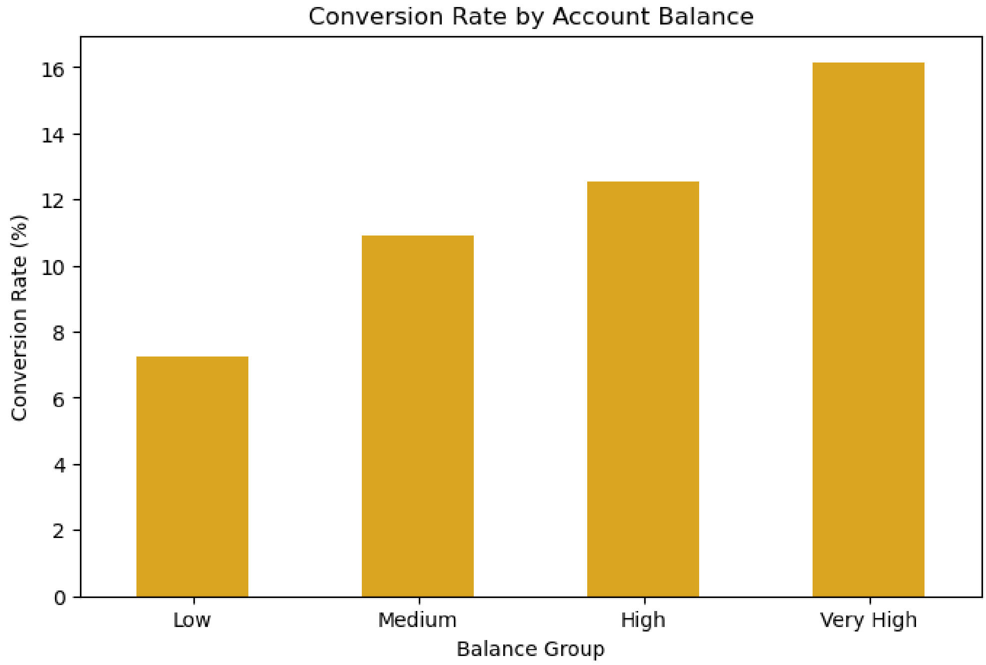

Figure 9: Conversion Rate by Account Balance Band

Key Observations

- Higher-balance customers convert at a materially higher rate.
- Low-balance customers rarely subscribe.
- Balance is one of the strongest financial predictors observed in the dataset.

Business Interpretation

Account balance is a strong pre-qualification signal. Prioritising outreach to medium- and high-balance customers, and de-prioritising low-balance customers, will directly reduce cost per acquisition.

10.9 Previous Campaign Outcome Analysis

Objective

Understand the value of prior engagement history in predicting future conversion.

Analysis

Conversion rates were computed for each value of the 'poutcome' attribute — success, failure, other and unknown.

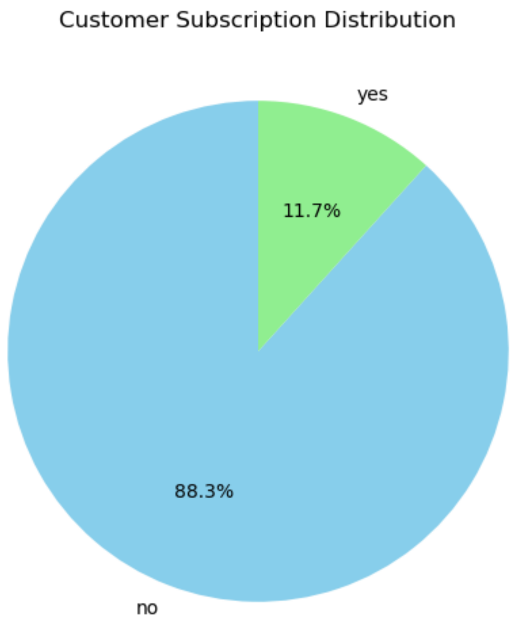

Figure 10: Conversion Rate by Previous Campaign Outcome

Key Observations

- Customers with a previous 'success' outcome convert dramatically higher than any other group.
- Even a previous 'failure' outcome outperforms customers with no prior contact.
- 'Unknown' history represents the majority of the base and converts near baseline.

Business Interpretation

Prior engagement is the single most powerful predictor identified in this study. A structured re-marketing programme aimed at previously successful customers is likely to be the highest-ROI initiative the bank can launch on the back of this analysis.

11. Key Findings

- The overall conversion rate is low, but performance varies sharply across segments.
- Cellular contact meaningfully outperforms fixed-line telephone contact.
- Campaign volume is concentrated in months that do not produce the strongest conversion.
- Students, retirees and tertiary-educated customers convert well above average.
- Single customers convert more strongly than married or divorced customers.
- The under-30 and over-60 age groups convert well above the middle-aged majority.
- Higher account balances are strongly associated with higher conversion.
- A prior successful campaign outcome is the single strongest predictor of future conversion.

12. Business Recommendations

12.1 Prioritise High-Converting Contact Methods

Consolidate outreach on cellular channels and phase out low-yield fixed-line campaigns. Complement calls with permission-based SMS follow-ups.

12.2 Optimise Campaign Timing

Rebalance campaign volume away from May and toward historically high-converting months such as March, September, October and December.

12.3 Improve Customer Targeting

Build a simple pre-qualification score using age band, account balance, job category and education level to route only high-propensity customers into active outreach.

12.4 Re-engage Positive Prior Customers

Launch a dedicated re-marketing workflow for customers with a previous 'success' outcome — this is the single highest-ROI action identified.

12.5 Adopt Data-Driven Marketing

Embed conversion tracking into every campaign, review results monthly, and iterate the targeting model as new outcomes become available.

13. Business Impact

- Higher customer conversion through segment-first targeting.
- Lower cost per acquisition by removing low-yield outreach.
- Improved customer experience by reducing over-contact of low-propensity customers.
- Better alignment between marketing spend and revenue outcomes.
- A repeatable analytical framework the bank can apply to future campaigns.

14. Challenges Faced

- Significant class imbalance in the target variable required careful interpretation of rates.
- Presence of 'unknown' categorical values needed thoughtful handling to avoid distortion.
- Communicating statistical findings in a business-friendly narrative required additional iteration.
- The dataset does not contain revenue-per-conversion, so ROI is inferred qualitatively.

15. Future Scope

- Build a supervised classification model (Logistic Regression, Random Forest, XGBoost) to score conversion propensity.
- Develop an interactive Power BI or Streamlit dashboard for marketing stakeholders.
- Extend the analysis with customer lifetime value and churn modelling.
- Automate monthly reporting through a scheduled pipeline.
- Incorporate A/B testing to validate recommendations in production campaigns.

16. Project Conclusion

This project demonstrates how a structured analytical approach can turn a large but raw marketing dataset into a focused set of business recommendations. By combining conversion-rate analysis across contact channels, timing and customer demographics with the strong signal contained in prior campaign outcomes, the bank has a clear path to increase subscription rates while lowering marketing cost. Equally importantly, the workflow developed here — data cleaning, EDA, segment-level conversion analysis, visualization and business interpretation — is directly reusable for future campaigns and future products.

17. References

- Moro, S., Rita, P., & Cortez, P. (2014). Bank Marketing Data Set. UCI Machine Learning Repository.
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